

Sarvar Abdullaev, PhD

sarrtv@gmail.com ◇ +1-307-346-6515 ◇ s-abdullaev.github.io

EDUCATION

- **King's College London** UK
PhD in Computer Science Sep 2011 - June 2016
Thesis: Pricing options via double auctions covering automated trading and multi-agent simulation of markets.
- **Dongguk University** South Korea
MSc in Computer Science, GPA 4.38/4.5 Mar 2007 - June 2009
Thesis: NLP-based web crawler that indexes conferences and workshops worldwide

SKILLS SUMMARY

- **Languages:** Python, TypeScript/JavaScript, PHP, Java, C#, C++, SQL, NetLogo, Prolog
- **Frameworks:** FastAPI, Flask, CrewAI, LangGraph, LlamaIndex, PyTorch, Angular, Laravel, Aiogram, Vue
- **DevOps:** Azure, Docker, K8s, Github Actions, Gitlab, RabbitMQ, Celery, Ansible, Prometheus, Graphana
- **Niche Tools:** Mosquitto, Influx, Fusion 360, PDDL, AMPL, Gurobi
- **Mathematics:** Calculus, Probability and Statistics, Linear Algebra, Optimization, Discrete Math, Game Theory

EXPERIENCE

- **Amudar.IO** Uzbekistan
Co-founder, CTO Jan 2020 - Present
 - **Created IoT ecosystem:** Deployed 150+ weather stations and pheromone traps across Uzbekistan to collect climate and image data. *Tech: Influx, Mosquitto, Docker, Laravel, Angular, PyTorch, MySQL*
 - **Built ML models:** Designed local deep learning models to count moths in pheromone traps and used it as a reward model for climate-based pest development prediction model. *Tech: OpenCV, TinyML, YOLOv4, Roboflow, RaspPi Zero*
- **Inha University in Tashkent** Uzbekistan
Associate Professor Sep 2016 - Present
 - **Teach courses:** Game Theory, Algorithms, Databases, Interaction Design and Introduction to Computer Science
 - **Founded labs:** Founded 3D printing and VR/AR labs at the university campus, organized public workshops and hackathons attracting more than 300+ participants
 - **Supervised projects:** Helped students in developing their startup ideas, e.g. Tilmoch raised \$300k for LLM based machine translation of Uzbek and 6 more other low resource languages.
- **Exelare** Remote (USA)
Software Developer Sep 2008 - Mar 2023
 - **Developed ATS from scratch:** For 15 years of work at Exelare as a software engineer, I designed and maintained Exelare ATS that automated the workflow for more than 2k recruiters by processing 20k+ resumes per day and using NLP to match skills to job descriptions. *Tech: C#, MS SQL, ASP.NET, Angular, RabbitMQ, Telerik, Sovren, other obsolete stuff*
 - **Integrated with resume banks and job posting portals:** Integrated Exelare ATS with portals such as Monster, LinkedIn, CareerBuilder, Dice, Indeed, Google Jobs, etc

PROJECTS

- **Wildfire Risk Assessment using Foundation Models (Geospatial Reasoning, Agentic AI):** (Work in progress)
Training VLM decoder using Google's PDFM and AlphaEarth embeddings to predict wildfire risk and socioeconomic impact. Training data is generated from WyoFire Monte-Carlo based simulation outputs.
- **Urban Air Quality Interpolation (Geospatial Analysis, IoT):** Deployed 10 air-quality monitoring stations across Tashkent, Uzbekistan. Built interpolation models using traffic and meteorological data to estimate PM1.0/2.5/10.0 concentrations.
- **Energy-Efficient Greenhouse Control (Time-series forecasting, Optimization, IoT):** Analyzed solar radiation, temperature, humidity, and ventilation patterns from IoT system to generate optimized heating schedules and to reduce energy consumption in greenhouse operations.
- **Vegetation Monitoring using UAV Imagery (Remote Sensing, Geospatial Analysis):** Designed the national web portal for near-real-time monitoring of vegetation-cover change using satellite (Landsat 8/9, Sentinel-2) and UAV images
- **Ebola Spread via Flights (Network Epidemiology, Agent-based Simulation):** Built real-time simulator forecasting disease spread using SIR epidemiological models across global airport networks. TechCrunch Disrupt 2014 winner.

PUBLICATIONS

- Pricing Options with Portfolio-holding Trading Agents in Direct Double Auction, *Frontiers in Artificial Intelligence and Applications* 285, pp. 1754–1755, 2016
- Emergence of Option Prices in Markets Populated by Portfolio-Holders, *In Proc. of 2nd European Workshop on Chance Discovery and Data Synthesis*, Hague, Netherlands, September 2016
- Direct Exchange Mechanisms for Option Pricing, *Lecture Notes in Computer Science* 8953, pp. 269–284, 2015
- An Optimization of CDN using Efficient Load Distribution and RADS Caching Algorithm, *Journal of Universal Computer Science*, Vol. 14, No. 14, pp. 2329-2342, 2008